MARIA HABIB

RESEARCH SOFTWARE ENGINEERING | AI EVALUATION & RELIABILITY INTERESTS

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RESEARCH PROFILE

Early-career software engineering researcher focused on evaluation reliability, robustness, false-positive analysis, and interpretable evidence in automated code analysis. Independently designed controlled experiments, compared baselines, and evaluated structural matching on external datasets. Seeking to apply these research methods to technical AI safety, especially trustworthy evaluations and understanding where automated metrics fail.

RESEARCH STRENGTHS

Empirical evaluation and benchmark design • Controlled transformations and negative controls • False-positive analysis • Human-in-the-loop decision support • Reproducible experiments • Python research tooling • Program analysis and testing

SELECTED INDEPENDENT RESEARCH

SimProbe — When Structure-Aware Matching Improves Java Similarity | Independent research

- Designed an experimental comparison of normalized-token similarity (C0) and method-aware one-to-one matching (C1), testing robustness to method reordering, class splitting, and renaming.
- Evaluated a frozen six-stratum sample of 6,000 BigCloneBench V2 pairs; 5,755 were measurable and 245 adapter failures were retained and reported.
- Reported an important boundary condition: method-aware matching helped on multi-method reorganizations but offered almost no advantage on single-method external fragments.
- Used deterministic sampling, frozen protocols, regression checks, provenance records, and SHA-256 artifact hashes to support reproducibility.

BehavClone — Evidence-Oriented Java Submission Similarity | Independent research prototype

- Built a Python/Tree-sitter pipeline to extract methods and constructors, compare multiple code representations using LCS, and align fragments with exact one-to-one matching.
- Separated structural similarity, cohort context, and externally imported behavioral test evidence instead of collapsing them into an automatic plagiarism verdict.
- Designed false-positive stress tests: in a 16-submission cohort, 48 of 96 unrelated cross-family pairs also reached normalized similarity 1.0, exposing the risks of over-normalization.
- Compared scores with JPlag 6.3.0 on 105 pairs from an authentic non-plagiarized Java cohort; explicitly documented limits of single-fragment validation.

CodeSpectra — Multi-Stage Academic Code Analysis Platform | Final-year project; team project

- Co-developed a full-stack research system for Type-1 to Type-4 code-similarity analysis using React/Vite, Node.js/Express, Python/FastAPI, PostgreSQL, structural analysis, and behavioral evidence.
- Built evidence views and instructor-facing review features; the design retains educator judgment rather than automatically declaring misconduct.
- Project discussed in Ptidej Research Blog; shared on Hacker News.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, C++ | Analysis: Tree-sitter, AST/token representations, LCS/edit similarity, matching, Random Forest | Engineering: Git/GitHub, GitHub Actions, REST APIs, FastAPI, Node.js/Express, React, PostgreSQL, regression testing, Ruff, SonarQube, Codacy

RESEARCH AND PROFESSIONAL EXPERIENCE

Digital Twin Research Intern — IoT Digital Twin Project | Jun–Aug 2025

- Investigated software quality and maintainability issues in Java-based Eclipse Ditto components using SonarQube, Codacy, source code, and issue history.

Software Engineer Intern — Spacebar Technologies | Sep–Nov 2025

- Contributed to React application development, integration, authentication-related features, testing, and debugging in a Git-based Agile workflow.

Teaching Assistant — PUCIT, University of the Punjab | Jan 2025–Jun 2026

- Supported Analysis of Algorithms, Computer Networks, and Probability & Statistics; evaluated 100+ student submissions and prepared structured feedback.

EDUCATION

BS Information Technology — PUCIT, University of the Punjab | 2022–2026 | CGPA 3.52/4.0

SELECTED WRITING AND PUBLIC WORK

- "CodeSpectra: The Illusion of Easy Coding — Why AI Still Demands Effort," Ptidej Research Blog (2026). https://blog.ptidej.net/codespectra-the-illusion-of-easy-coding-why-ai-still-demands-effort/
- CodeSpectra shared on Hacker News: https://news.ycombinator.com/item?id=48162094 (community submission, not a peer-reviewed publication).
- Independent research manuscripts / technical reports: SimProbe; BehavClone. Research portfolio and code available through GitHub.

RESEARCH INTERESTS FOR AI SAFETY

Evaluation validity and failure modes; robustness versus discrimination; benchmark design and dataset shift; interpretability of automated evidence; reproducible safety-relevant software experiments. These are prospective AI safety directions, not claims of completed AI safety research.